

SENIOR PROJECT TH1-2025

Interpretable AI for Meditative State Classification: Visualizing Brainwave-Based Predictions and Personalized Insights

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Abstract

This project details the design and prototype implementation of an Interpretable AI system for classifying meditative states using the Muse 2 EEG headband. The system addresses the lack of objective feedback in meditation practice by classifying brainwaves into four distinct levels: Baseline, Physical Relaxation (Alpha), Mental Focus (Theta), and Deep Meditation (Delta). To establish objective, data-driven ground truth for these levels, our architecture first employs unsupervised clustering on a validated public dataset of EEG recordings from experienced meditators (Thai Buddhist monks). Following this baseline definition, the pipeline utilizes BrainFlow for real-time Muse 2 data acquisition, spectral feature engineering, and a Support Vector Machine (SVM) classifier with subject-independent cross-validation and SMOTE class balancing. This document presents the system design, the verified software prototype using both synthetic data injection and dataset clustering, the full two-phase SVM training report with 95% confidence intervals, and live inference results from real Muse 2 sessions.

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Chapter 1

Project Concept

In recent years, meditation has gained increasing attention as a practice that promotes mental well-being, emotional balance, and cognitive clarity [1]. However, many individuals who meditate regularly struggle with the lack of feedback or guidance regarding their progress. Technologies such as consumer-grade EEG headbands have made it possible to capture and analyze brain activity in real time with reasonable accuracy [2]. These advancements open the door for integrating artificial intelligence with meditation practice, allowing for objective measurement, progress tracking, and personalized guidance. This project explores how brain-sensing technology and AI can be combined to better understand meditative states and support practitioners in their journey.

1.1 Summary

This project focuses on developing an AI-powered system that classifies and interprets brainwave signals collected from the Muse 2 brain-sensing headband. By analyzing electroencephalography (EEG) data, the system aims to identify different stages of meditative preparation and practice. The approach integrates data preprocessing, noise reduction, and machine learning model development to provide both accurate classification and interpretable visual insights. Ultimately, the system is designed to help meditation practitioners monitor their progress and receive personalized feedback. A core component of this work is exploratory and statistical data analysis of EEG signals to examine their spectral characteristics across meditation states. The insights gained from this analysis directly inform feature selection, clustering strategies, and model design, ensuring that the overall classification pipeline is grounded in empirical evidence rather than heuristic assumptions.

1.2 Motivation

Many people want to meditate daily but struggle because their progress feels unknown. Good teachers can be rare or expensive to find, so many end up practicing without guidance and feeling that their progress is stagnant. Getting an answer to a simple question like ‘Am I doing this right?’ can feel so cumbersome. Unlike physical exercise, where measurable indicators such as distance or strength can show improvement, meditation progress is often subjective and difficult to assess without the guidance of an experienced teacher. This creates a barrier for practitioners who wish to deepen their practice but have limited access to expert supervision. Our project aims to solve this problem by creating a digital tool to help meditators understand themselves. It will provide a clear baseline or guideline for their practice. With advancements in wearable EEG devices such as Muse 2, it is now possible to capture brain signals in real time [2]. This creates an opportunity to apply AI techniques to provide objective, data-driven insights into meditation states, supporting both beginners and experienced meditators.

1.3 Users and Benefits

1.3.1 Users

The platform is designed for a diverse audience, each with specific goals:

- **Beginner Meditators:** Newcomers who want clear and actionable guidance to help them build a consistent meditation habit.
- **Intermediate Practitioners:** Individuals with an established practice who are looking for measurable indicators to track and validate their progress.
- **Advanced Meditators:** Experienced practitioners interested in using granular data for the fine-tuning of specific mental states.
- **Researchers:** Academics and neuroscientists who can use the platform as a tool for exploring the correlations between brainwave patterns and meditative states.

1.3.2 Benefits

The core value of the tool is to make meditation progress tangible and accessible. Key benefits include:

- **Personalized Feedback:** Provides real-time or post-session feedback on meditation depth, offering personalized insights tailored to the user’s unique brainwave activity.

- **Cost-Effectiveness:** Reduces the reliance on expensive one-on-one sessions with experts, making deep practice guidance more accessible.
- **Motivation through Visibility:** Encourages consistency by offering visible progress indicators. Seeing tangible improvement over time serves as a powerful motivator to continue the practice.
- **Long-Term Growth Tracking:** Allows each user to build a personal meditation baseline, creating a clear and objective benchmark for tracking long-term growth and self-awareness.

1.4 Typical Usage

Here is how a person would typically use the system:

- **Step 1: Get Ready to Meditate:** First, the user puts on the Muse headset. Then, they start the Muse-LSL app on a computer or phone. This app connects to the headset and starts sending the raw brainwave data.
- **Step 2: Start the App:** Next, the user opens our main app. This app, made with the BrainFlow library, finds and connects to the data stream from the headset. As the user meditates, our app gets the brainwave signals live. It uses BrainFlow's tools to clean up the signals by filtering out noise.
- **Step 3: AI Analysis and Feedback:** The system sorts the clean data into 5-min blocks. Each block is then sent to our AI model. The model looks at the brainwave patterns to figure out the user's meditation level (for example, if they are relaxed or focused). The app then gives the user simple feedback right away, like a soft sound or a text change on the screen.
- **Step 4: Check Your Results:** When the meditation is over, the user can see a report. This report shows how they did, like how long they stayed in each meditation level. This helps them track their progress over time.

1.5 Main Challenges

The main challenges in processing brain wave data include:

- **Data Acquisition and Quality:** The challenge is capturing clean EEG data from consumer devices which are prone to noise, where the noise and signal are mixed with artifacts (e.g., blinking, jaw movement) [3]. Effective filtering and strong API integration are required for reliable signals.

- **Data Processing and Feature Engineering:** Raw EEG data is complex. The challenge is extracting meaningful features by segmenting sessions (e.g., 1–5 minutes) and applying signal processing techniques like FFT to analyze brainwave frequencies (Alpha, Beta, Theta, Delta), which correlate with mental states [4].
- **Defining Objective Meditation States:** A major challenge is mapping subjective experiences like “deep focus” or “mind-wandering” to measurable brainwave patterns. This requires collaboration between neuroscientists and meditation teachers to build a standardized framework and labeled ground-truth dataset [5].
- **Model Development and Classification:** The core task is training machine learning models (e.g., CNN, RNN) to classify signals into meditation levels. Achieving accuracy and generalization across individuals remains difficult.

Chapter 2

Requirements Specification

2.1 System Description

This project aims to develop an AI powered system that analyzes brainwave signals from the Muse 2 brain-sensing headband to classify meditation states and provide personalized feedback. By processing raw EEG data along with supporting signals (heart rate, motion), the system applies signal preprocessing, feature extraction, and machine learning to identify meditation depth and quality.

Unlike traditional meditation practice, where progress is often subjective and hard to measure, this system provides objective, data-driven insights. The platform supports users across all experience levels, from beginners seeking guidance, to advanced meditators refining their practice, and researchers exploring the neuroscience of meditation.

Users wear the Muse 2 headset, which streams brainwave data into the application. After cleaning the signals, the AI model evaluates meditation levels. By combining wearable technology, machine learning, and interpretability, the system helps users track their meditation progress, stay motivated, and develop a deeper awareness of their mental states.

This design specification was refined following exploratory analysis of an external meditation EEG dataset. The findings informed feature selection, spectral analysis parameters, and model design choices.

2.1.1 Product Perspective

The system is a standalone, proof-of-concept application designed for both personal use and research. It operates independently and does not require integration with other enterprise systems.

- **System Inputs:** The system's primary inputs are live data streams from the Muse 2 headband, including 4-channel EEG, PPG (for heart rate), and IMU (for motion) data.

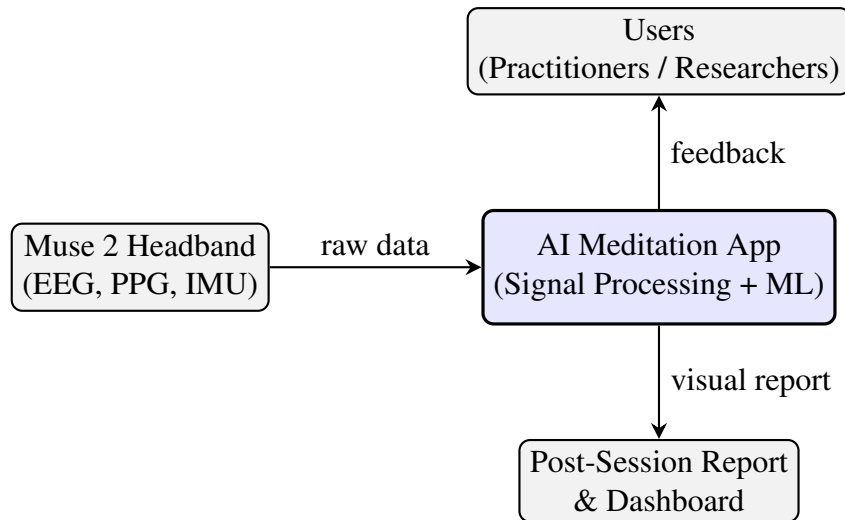


Figure 2.1: System context: product perspective showing inputs, outputs, and users.

- **System Outputs:** The primary output is a post-session visual analytics report that summarizes the user's meditative states, stillness, and heart rate variability over the session's duration.
- **Users:** The intended users are beginner-to-intermediate meditation practitioners seeking objective feedback, as well as researchers studying the neuro-correlates of meditation.

2.1.2 Product Functions

The system will perform the following five core functions to transform raw brainwave data into actionable insights:

1. **Signal Acquisition & Ingestion:** The system will establish a connection with the Muse 2 headband via Bluetooth and use the BrainFlow library to create a stable data stream. It will simultaneously record EEG, PPG, and IMU signals with unified timestamps.
2. **Data Preprocessing:** The system will apply digital signal processing techniques to the raw data. This includes applying band-pass and notch filters to remove noise and using the IMU data to detect and flag segments contaminated by physical motion artifacts (e.g., blinks, jaw clenches).
3. **Feature Engineering:** From the cleaned signals, the system will extract a set of meaningful numerical features. These include spectral features (power in Alpha, Beta, Theta bands), spectral ratios, and regional spatial aggregates (frontal alpha asymmetry, frontal theta mean, occipital alpha mean).

4. **Machine Learning Classification:** The system will use the extracted features as input for machine learning models. It will be capable of training and evaluating a Support Vector Machine (SVM) with subject-independent cross-validation (Group-KFold) and SMOTE class balancing to classify each window of data into a discrete meditative state.
5. **Post-Session Analytics & Visualization:** After a session is complete, the system will generate and display a user-friendly dashboard. This report will visualize the predicted states over time on a timeline, show the overall percentage of time spent in each state, model confidence distribution, and band power trends.

2.1.3 Data-Driven Design Considerations

The overall system design and functional requirements were informed by exploratory and statistical analysis of real-world meditation EEG data obtained from an external benchmark dataset. This analysis was conducted to ensure that architectural and algorithmic decisions were grounded in observable signal characteristics rather than heuristic assumptions. The outcomes of this exploratory analysis are presented in Design Specification 5.

Specifically, data analysis contributed to the following design decisions:

- Spectral analysis of EEG signals revealed distinguishable patterns in delta, theta, alpha, and beta frequency bands across different mental states, motivating the inclusion of frequency-domain feature extraction using Power Spectral Density (PSD).
- Preliminary clustering and visualization of band power distributions indicated natural groupings in the data, supporting the selection of multiple discrete meditation states rather than a binary classification approach.
- Comparative analysis between meditative and non-meditative conditions demonstrated partial separability in feature space, justifying the use of both baseline feature-based classifiers and regularized ensemble models to capture discriminative spectral patterns while avoiding overfitting.

These insights directly influenced the definition of system functions, feature extraction requirements, and the structure of the machine learning pipeline described in subsequent sections.

2.2 Requirements

Data Acquisition & Ingestion

- DA-1:** The system **shall** interface with the Muse 2 brain-sensing headband via Bluetooth Low Energy (BLE).

DA-2: The system **shall** use the BrainFlow library to ingest data streams from the Muse 2.

DA-3: The system **shall** capture the following signals: 4-channel EEG (TP9, AF7, AF8, TP10) at 256 Hz, PPG for Heart Rate (HR) and Heart Rate Variability (HRV), and IMU for motion data.

DA-4: The system **shall** record and unify all data streams with monotonic timestamps.

Data Preprocessing

DP-1: The system **shall** apply a band-pass filter from 1–45 Hz to the raw EEG signal.

DP-2: The system **shall** apply a 50 Hz notch filter to remove electrical interference.

DP-3: The system **shall** use IMU data to detect and flag segments contaminated with motion artifacts.

DP-4: The system **shall** segment the continuous data into 2.0-second windows with a 50% overlap.

Feature Extraction

FE-1: The system **shall** calculate Welch’s Power Spectral Density (PSD) for Delta, Theta, Alpha, Beta, and Gamma bands for each window. This requirement is supported by preliminary analysis on real-world meditation EEG data, which demonstrated robust discrimination between mental states using PSD-based band power features.

FE-2: The system **shall** compute key spectral ratios, including Theta/Alpha and Alpha/Beta. These ratios were selected based on observed differences in alpha and theta activity between meditation and thinking states.

FE-3: The system **shall** extract regional spatial features including frontal alpha asymmetry (FAA), frontal theta mean, and occipital alpha mean.

FE-4: The system **shall** engineer a total of 19 subject-transferable features fully reproducible from the Muse 2’s 4-channel CSV at inference time, using Muse-proximity channel selection to reduce the domain gap between the 64-channel training dataset and the 4-channel inference device.

Machine Learning Model

- ML-1:** The system **shall** classify data into at least four distinct meditative states (e.g., {baseline, low, medium, high focus}).
- ML-2:** The system **shall** implement and evaluate an optimized Support Vector Machine (SVM) classifier (RBF kernel, $C=100$, $\gamma=0.001$) with subject-independent GroupKFold cross-validation.
- ML-3:** The system **shall** apply SMOTE oversampling within each CV fold to address class imbalance in the minority Deep Meditation class.
- ML-4:** The system **shall** provide a confidence score for each prediction.
- ML-5:** The system **shall** include logic to warn users when more than 20% of windows fall below a 0.50 confidence threshold, citing likely causes such as poor electrode contact or domain gap.

User Interface & Analytics

- UI-1:** The system **shall** present a post-session report to the user.
- UI-2:** The report **shall** include a timeline visualizing the predicted meditative states over the session.
- UI-3:** The report **shall** display summary statistics, including percentage of time in each state, dominant state, and confidence distribution.
- UI-4:** The system **shall** allow users to track their progress across multiple sessions.

Performance

- PM-1:** The classifier **must** achieve a subject-independent CV accuracy of ≥ 0.85 on the monk benchmark dataset.
- PM-2:** The macro-F1 score **must** exceed 0.88 across all four meditation levels.
- PM-3:** After temporal smoothing, the final output **should** exhibit ≤ 1 state transition per minute during steady meditation blocks.

Reliability & Robustness

- RR-1:** The system **must** correctly identify and abstain from classifying at least 95% of windows with high motion or poor signal quality.

RR-2: The model **must** be trained and validated using a strict subject-independent split to ensure generalizability.

RR-3: The training-to-CV accuracy gap **should** not exceed 10% (mild overfitting threshold).

Usability

UB-1: The post-session report **must** be clear, intuitive, and understandable to users with no background in neuroscience.

Chapter 3

Design Specification

This chapter describes the internal design of the system. It details the architectural components and the underlying methodologies chosen to fulfill the requirements outlined in Chapter 2. For each key design decision, a rationale is provided to justify the chosen approach.

3.1 System Architecture

The system is designed as a modular, data-driven pipeline that processes biosensor data in real-time to classify meditative states. The architecture is divided into distinct, loosely-coupled modules, each responsible for a specific stage of the data processing workflow. This modularity allows independent development, testing, and potential future upgrades of each component.

The high-level architecture, illustrated in Figure 3.1, shows the flow of data from the user's Muse 2 headband, through various processing stages, to the final user interface. The core of the system involves an offline training phase where the machine learning model is built and a real-time inference phase where the trained model is deployed to provide feedback to the user.

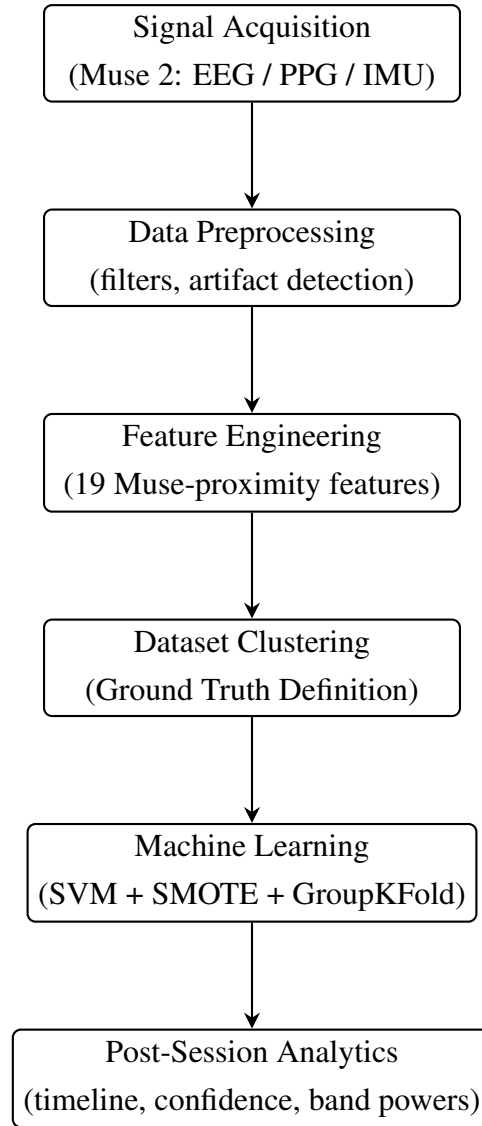


Figure 3.1: System Architecture: from signal acquisition to post-session analytics, including offline clustering for ground-truth definition and an SVM with subject-independent validation.

The primary components of the system are:

- **Data Acquisition Layer:** The physical hardware (Muse 2) that captures raw biosignals.
- **Data Ingestion & Streaming Module:** A software layer that establishes a connection with the hardware and streams data into the system.
- **Signal Preprocessing Module:** Cleans and prepares the raw data for analysis by filtering noise and handling artifacts.
- **Feature Extraction Module:** Transforms the cleaned signal into 19 subject-transferable numerical features using Muse-proximity channel selection to minimize domain gap between training and inference.

- **Ground Truth Definition Module (Offline):** An unsupervised machine learning pipeline applied to a benchmark dataset of experienced meditators. This module mathematically isolates and defines the target meditation states, serving as the foundational baseline for the Machine Learning Core.
- **Machine Learning Core:** Contains the trained SVM model responsible for classifying the feature set into a meditative state, with subject-independent GroupKFold validation and SMOTE class balancing. It has both offline training and real-time inference capability.
- **Data Visualization & User Interface:** The front-end application that presents the classification results, confidence scores, and session analytics to the user.

3.2 System Execution Walkthrough

This section demonstrates how the implemented system processes EEG data from input acquisition to final user feedback.

1. **Input Acquisition:** The system receives a continuous EEG stream sampled at 256 Hz, either from the synthetic generator or an external dataset. Each session is segmented into overlapping time windows of fixed duration.
2. **Signal Preprocessing:** Each window undergoes band-pass filtering and artifact detection to reduce noise and remove motion- or blink-related distortions.
3. **Feature Extraction:** For each window, the system computes a 19-dimensional feature vector, including spectral band powers, statistical descriptors, and cross-channel connectivity metrics.
4. **Inference and Classification:** The extracted features are passed to a trained SVM classifier, which outputs both a predicted meditation state and a probability distribution over all possible states.
5. **Post-Session Analysis and Feedback:** Classification results are aggregated over time to produce a session timeline, confidence metrics, and interpretable summaries such as dominant frequency bands.

3.3 Detailed Design

This section provides a low-level view of each component identified in the System Architecture.

3.3.1 Data Acquisition Layer

The system will utilize the Muse 2 headband as its primary data acquisition device. This choice is justified by its status as a cost-effective consumer-grade device that has been validated for research purposes [2]. Critically, its multimodal sensor suite provides the rich data necessary for robust analysis. The system will capture data from three types of sensors simultaneously:

- **Electroencephalography (EEG):** 4 channels (TP9, AF7, AF8, TP10) will be recorded at a sampling rate of 256 Hz. This is the primary signal for meditative state classification.
- **Photoplethysmography (PPG):** This sensor data will be used to derive heart rate (HR) and heart rate variability (HRV).
- **Inertial Measurement Unit (IMU):** An accelerometer and gyroscope will be used to detect head movement.

While EEG is the core signal, the IMU and PPG data are essential for enhancing the system’s robustness. The IMU data are crucial for artifact detection and rejection (e.g., flagging data corrupted by head movement), while HRV provides additional physiological context related to the user’s autonomic nervous system response.

3.3.2 Data Ingestion & Streaming Module

This module will employ the BrainFlow library in Python to establish a connection to Muse 2 via Bluetooth Low Energy (BLE). BrainFlow was chosen because it provides a powerful, unified API that abstracts away the complexities of connecting to various biosensors, simplifying development and making the application less dependent on specific hardware details. Once connected, BrainFlow will stream the data into a lab streaming layer (LSL) outlet. A separate Python script will then create an LSL inlet to record the data for processing. The use of LSL is a key design decision; it is the industry standard for real-time, synchronized streaming of time-series data. This decouples data acquisition from data processing, allowing them to run in separate processes for a more scalable and robust architecture, while ensuring all data streams are precisely timestamped.

3.3.3 Ground Truth Definition via Unsupervised Clustering

Before supervised models can accurately classify live data from the Muse 2, the system must first establish an objective definition of what constitutes different meditative states. To achieve this, an offline unsupervised learning phase is conducted using a validated public

dataset: *EEG absolute and relative powers during mindfulness meditation: Data from Thai Buddhist monks* (Mendeley Data) [6].

This benchmark dataset contains high-quality, 64-channel EEG recordings of experienced monks during 30-minute sessions. By leveraging this data, the system grounds its classifications in the neurological baselines of expert meditators rather than arbitrary thresholds. The unsupervised pipeline consists of the following steps:

1. **Data Restructuring and Normalization:** The raw EEG power data is pivoted for time-series continuity and normalized to ensure that extreme variances in specific frequency bands do not skew the distance calculations during clustering.
2. **Unsupervised Clustering:** We apply clustering algorithms (evaluating $K = 2$ through $K = 6$ clusters) to the normalized dataset to group natural neural activity patterns without prior labels.
3. **Cluster Evaluation:** The optimal number of clusters is determined using evaluation metrics. The Elbow Method and Silhouette Scores are analyzed to mathematically verify the optimal separation of brainwave states.
4. **State Mapping:** Once the optimal clusters are identified, dimensionality reduction techniques like UMAP are used to visualize the separation. The resulting clusters are then mapped to our target states based on their dominant spectral power (e.g., a cluster showing high Delta power is mapped to “Deep Meditation”). These defined centroids and distributions become the ground-truth targets for the supervised models.

3.3.4 Signal Preprocessing Module

Preliminary experimentation revealed that Muse 2 CSV logs may contain missing values, non-numeric samples, and segments corrupted by poor electrode contact or head motion. Therefore, the inference pipeline includes explicit validation and quality gating prior to feature extraction to ensure numerical stability and to reduce artifact-driven misclassification.

Unlike traditional pipelines that compute PSD from raw EEG, this system uses the Muse 2’s **pre-computed band power streams** (Delta/Theta/Alpha/Beta/Gamma) provided in the CSV. As a result, preprocessing focuses on **data cleaning and window-level exclusion** rather than time-domain filtering.

Each recording is processed using a sliding-window scheme, where each 2-second window undergoes the following steps:

1. **Numeric validation:** Band power columns and IMU values are coerced to numeric format. Invalid values are treated as missing and handled safely to prevent numerical errors in feature computation.

2. **Signal-quality gating (Muse contact indicators):** Samples are flagged as low quality if `HeadBandOn` $\neq 1$ or if any HSI channel indicates poor contact ($\text{HSI} > 2$). A window is discarded if fewer than 70% of its samples pass quality checks.
3. **Motion artifact gating (IMU):** The standard deviation of the resultant acceleration magnitude is computed per window. Windows with motion variability exceeding 0.08 g are excluded to reduce muscle and movement artifacts.
4. **Windowing:** Valid data are segmented into 2-second windows with a 1-second stride (50% overlap). This window length captures low-frequency dynamics while overlap provides smoother temporal state transitions.

3.3.5 Feature Extraction Module

The goal of this module is to reduce the dimensionality of the preprocessed data while retaining the information most discriminative of meditative states. A critical design revision addresses the domain gap between the 64-channel monk training dataset and the 4-channel Muse 2 inference device. Training features are computed using only the subset of monk dataset channels whose 10–20 system positions are spatially closest to the four Muse 2 electrode locations:

- **TP9** (left temporal-parietal): matched to {TP7, T7, TP9, P7, P9}
- **AF7** (left frontal): matched to {AF3, F7, AF7, Fp1, F5, F3}
- **AF8** (right frontal): matched to {AF4, F8, AF8, Fp2, F6, F4}
- **TP10** (right temporal-parietal): matched to {TP8, T8, TP10, P8, P10}

From these Muse-proximity channels, 19 subject-transferable features are engineered across four groups. For each 2-second window, the Muse 2 band power values (exported in Bels, i.e. $\log_{10}$ scale) are first converted to linear power before relative power is computed:

$$P_{\text{linear}}(ch, \text{Band}) = 10^{P_{\text{Bels}}(ch, \text{Band})}, \quad P_{\text{rel}}(ch, \text{Band}) = \frac{P_{\text{linear}}(ch, \text{Band})}{\sum_b P_{\text{linear}}(ch, b)}$$

The 19 features are then:

- **Group A — Mean Relative Power (5 features):** Mean relative power across Muse-proximity channels for each of the five bands ($\delta, \theta, \alpha, \beta, \gamma$).
- **Group B — Spatial Standard Deviation (5 features):** Standard deviation of relative power across the selected channels per band, measuring spatial heterogeneity.

- **Group C — Spectral Ratios (6 features):** Pairwise ratios θ/α , α/β , δ/α , θ/β , δ/β , γ/α . These are more robust to inter-subject amplitude differences than absolute power alone.
- **Group D — Regional Spatial Features (3 features):** Frontal alpha asymmetry (FAA = left frontal α – right frontal α), frontal theta mean (average of AF7/AF8 theta), and occipital alpha mean (average of TP9/TP10 alpha, the Muse 2 channels nearest to posterior alpha generators).

All 19 features are identical between training and inference, ensuring full reproducibility from the Muse 2’s 4-channel CSV.

3.3.6 Machine Learning Pipeline

Frequency-domain analysis demonstrated that meditation-related EEG signals exhibit separable spectral characteristics, particularly in alpha and theta bands. This supports the use of feature-based classifiers such as a Support Vector Machine (SVM) as the primary model, with deliberate regularization to prevent memorization of the small 341-sample dataset.

1. Model Architecture An SVM with an RBF kernel is used. To prevent overfitting while preserving discriminative power, the following hyperparameters are applied via cross-validated grid search optimizations:

`kernel='rbf', C=100, gamma=0.001, and class_weight='balanced'.`

SMOTE (Synthetic Minority Oversampling Technique) is applied within each CV fold on the training portion only to address the severe imbalance in Level 3 (Deep Meditation), which represents only 27 of 341 samples (8%). Synthetic samples are generated by interpolation between existing minority-class samples and their k nearest neighbors: $x_{\text{syn}} = x_i + \lambda(x_{nn} - x_i)$, $\lambda \sim \mathcal{U}(0, 1)$.

A label circularity diagnostic is run using a depth-1 decision tree. If the depth-1 accuracy exceeds 0.70, the labels are likely trivially derivable from a single band-power threshold, indicating circular mapping rather than genuine learned patterns.

2. Classification Levels The model predicts four discrete meditation states strictly aligned with the clusters isolated during the unsupervised evaluation of the benchmark dataset (detailed in Section 3.2.5):

- **Level 0 (Baseline):** Active, distracted state characterized by high Beta/Gamma activity, aligning with the non-meditative baseline cluster.
- **Level 1 (Relaxed):** Physical calmness with dominant Alpha activity (8–12 Hz), mapping to the initial transition cluster.

- **Level 2 (Focused):** Mental concentration with prominent Theta activity (4–8 Hz), representing sustained meditative focus.
- **Level 3 (Deep):** Profound stillness characterized by Delta activity (1–4 Hz), mapped directly to the expert-level deep meditation cluster found in the Thai Buddhist monk dataset.

3. Validation Strategy To ensure generalizability, the models are evaluated using **Subject-Independent Cross-Validation (GroupKFold)** with 5 folds across 60 subjects, ensuring the system is tested on users it has never encountered during training. The 95% confidence interval is computed using the t -distribution over fold-level accuracy scores. The overall performance estimate is the pooled accuracy across all fold predictions.

3.3.7 Data Visualization & User Interface

The post-session dashboard presents three panels: a session timeline of predicted meditation levels color-coded by state, a model confidence plot with high ($\geq 70\%$) and low ($< 50\%$) thresholds, and a band power trend chart showing relative power per band over time. Summary statistics include total session duration, dominant state, time and percentage spent per level, mean and median confidence, and confidence tier distribution. This design translates complex neurophysiological data into easily digestible insights.

3.3.8 Objective vs. Current Implementation Workflow

The system architecture illustrated previously outlines the **objective workflow** for the final integrated application. In this ideal end-to-end pipeline, all stages operate sequentially on live data: Signal Acquisition (Muse 2: EEG / PPG / IMU) → Data Preprocessing (filters, artifact detection) → Feature Engineering (19 Muse-proximity features) → Dataset Clustering (Ground Truth Definition) → Machine Learning (SVM + SMOTE + GroupKFold) → Post-Session Analytics (timeline, confidence, band powers).

However, to establish a scientifically valid baseline, the **current prototype workflow** logically separates the training and inference pipelines:

- **Training Phase (Mendeley Dataset):** To train the SVM model, the system currently utilizes the Mendeley benchmark dataset [6]. Because this dataset consists of high-quality recordings from experienced monks that have already been rigorously preprocessed, our training pipeline bypasses the raw signal acquisition and initial artifact filtering steps. Instead, it directly performs dataset clustering (to define ground-truth states) and extracts the 19 Muse-proximity features to train the model.

- **Inference Phase (Live Muse 2 Data):** The full operational pipeline is actively applied during the testing and inference phase. Raw, real-world Muse 2 recordings (such as test sessions S46 and S47) undergo the complete sequence of raw data ingestion, signal quality gating, motion artifact exclusion, and feature engineering before being evaluated by the pre-trained SVM.

This bifurcated approach ensures that the model learns from pristine, validated neurological baselines while still proving the system’s capability to clean, process, and classify raw, noisy data from consumer hardware in a real-world setting.

3.4 System Use Cases and Operational Scenarios

This section illustrates how the implemented system operates under realistic scenarios, demonstrating its practical functionality, robustness, and interpretability.

3.4.1 Use Case 1: Offline Meditation Session Analysis

- **Objective:** To analyze a complete meditation session and quantify the user’s mental state progression over time.
- **Input:** A continuous EEG recording sampled at 256 Hz, obtained from either the synthetic data generator or an external dataset, representing a single meditation session.
- **Process:** The input signal is segmented into overlapping time windows and passed through the preprocessing pipeline to remove noise and artifacts. For each window, a 19-dimensional feature vector is extracted, incorporating spectral band powers, statistical descriptors, and cross-channel connectivity features. These features are then fed into the trained SVM classifier, which outputs a predicted meditation state and a probability distribution for each window.
- **Output:** The system produces a temporal profile of meditation states across the session, accompanied by confidence scores and summary statistics such as dominant frequency bands.
- **Outcome:** This use case demonstrates the system’s ability to transform raw EEG signals into interpretable insights, allowing users to observe transitions between mental states such as Focused, Relaxed, and Deep meditation.

3.4.2 Use Case 3: Live Inference on Muse 2 Sessions

- **Objective:** To demonstrate real-time classification of meditation states using live EEG data streamed from the Muse 2 headband.
- **Input:** A continuous EEG data stream acquired in real time from the Muse 2 device, sampled at 256 Hz during an active meditation session.
- **Process:** Incoming EEG data is buffered and segmented into sliding time windows. Each window is processed through the same preprocessing and feature extraction pipeline used during offline training. The extracted features are passed to the trained classification model, which performs real-time inference and outputs predicted meditation states along with associated confidence scores.
- **Output:** A live stream of meditation state predictions, confidence levels, and summary indicators such as dominant frequency bands, updated continuously throughout the session.
- **Outcome:** This use case validates the system's ability to operate under real-time constraints, providing immediate and interpretable feedback to users. It directly supports the live inference experiments presented in Section 4.7 and demonstrates the system's readiness for deployment in practical meditation settings.

Chapter 4

System Implementation and Prototype Results

4.1 Prototype Implementation Overview

The experiments and evaluations presented in this chapter correspond directly to the system use cases defined in Section 3.4. The core system has been implemented in Python (v3.9), utilizing the `scikit-learn` library for the machine learning engine, `imbalanced-learn` for SMOTE oversampling, `umap-learn` for dimensionality reduction, and `scipy` for signal processing and statistical analysis. To ensure scalability and maintainability, the software architecture follows a modular object-oriented design, structured into four primary operational modules:

Clustering Module (`clustering.py`): Analyzes the benchmark dataset using unsupervised learning (K-Means) to establish objective ground-truth labels for meditative states based on expert practitioner EEG data.

Training Module (`train_meditation_classifier.py`): This module manages the end-to-end pipeline of Muse-proximity channel selection, 19-feature engineering, SMOTE class balancing, SVM training, subject-independent GroupKFold evaluation with 95% confidence intervals, and serialization of the trained model artifacts (`.pkl` files).

Inference Module (`test_meditation_classifier.py`): Designed for deployment, this module loads the pre-trained model to perform inference on Muse 2 CSV recordings. It applies Bels-to-linear conversion, signal quality gating, motion gating, 19-feature extraction, per-window confidence scoring, and generates session timeline plots.

4.2 Dataset Clustering and State Evaluation

This section presents the core data analysis conducted to establish objective meditation state definitions. Using unsupervised clustering, spectral profiling, and dimensionality reduction

techniques, we analyze EEG power distributions to identify naturally occurring brainwave patterns among experienced meditators.

To establish objective baselines prior to training the classifier, an unsupervised clustering pipeline was applied to the Mendeley dataset. This process identifies intrinsic structure in the data without relying on subjective labels.

4.2.1 Cluster Optimization

Clustering was performed using the K-means algorithm, which partitions the dataset into K clusters by minimizing within-cluster variance. The objective function is defined as:

$$J = \sum_{i=1}^K \sum_{x \in C_i} \|x - \mu_i\|^2$$

where C_i represents cluster i and μ_i is its centroid. This formulation ensures that samples within the same cluster are as similar as possible in feature space.

To determine the optimal number of clusters, multiple values of K were evaluated using cluster validation metrics. In particular, the silhouette score was used to measure cluster separation:

$$S = \frac{b - a}{\max(a, b)}$$

where a is the average intra-cluster distance and b is the minimum inter-cluster distance to neighboring clusters. Higher values indicate better-defined and more separated clusters.

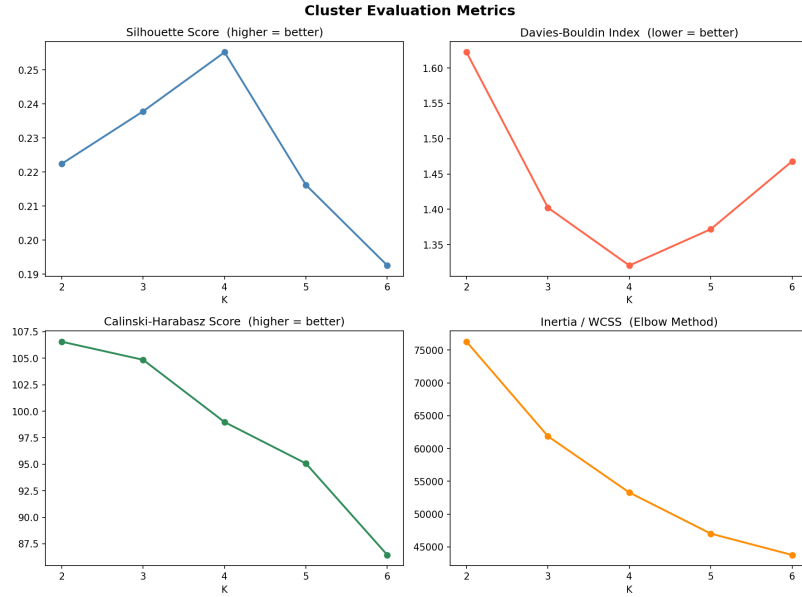


Figure 4.1: Cluster evaluation metrics determining the optimal number of meditative states. The analysis mathematically confirms $K = 4$ as the optimal separation.

As shown in Figure 4.1, the evaluation metrics consistently indicate that $K = 4$ is the optimal number of clusters. This aligns with the hypothesized four-stage meditation structure (Baseline, Relaxed, Focus, Deep), providing a data-driven justification for the classification framework.

4.2.2 State Mapping via Spectral Profiles

After clustering, each group is represented by its centroid:

$$\mu_i = \frac{1}{|C_i|} \sum_{x \in C_i} x$$

These centroids capture the average spectral profile of each cluster and serve as representative prototypes of the underlying brain states.

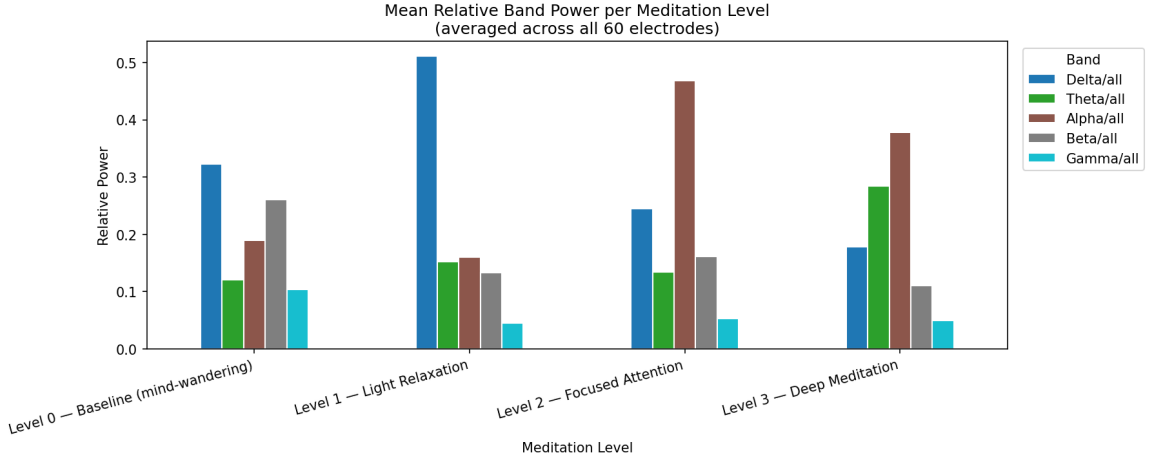


Figure 4.2: Spectral band profiles for each of the $K = 4$ clusters, illustrating the varying dominance of Delta, Theta, Alpha, Beta, and Gamma frequencies.

Figure 4.2 illustrates the spectral profiles of the clusters. By analyzing the relative power distribution across frequency bands, each cluster is mapped to a corresponding meditation state. Clusters with dominant Beta/Gamma activity correspond to the Baseline (active/distracted) state, Alpha-dominant clusters indicate Physical Relaxation, Theta-dominant clusters represent Mental Focus, and Delta-dominant clusters correspond to Deep Meditation.

These centroid-derived profiles provide a mathematically grounded and interpretable definition of meditation states, forming the objective ground truth for the supervised learning stage.

4.2.3 Dimensionality Reduction and Visualization

To visualize the high-dimensional EEG feature space, we employed Uniform Manifold Approximation and Projection (UMAP). UMAP constructs a low-dimensional embedding

by preserving local neighborhood structure from the high-dimensional space. Given a set of data points, UMAP optimizes the following objective:

$$\mathcal{L} = \sum_{i \neq j} \left[w_{ij} \log \frac{w_{ij}}{v_{ij}} + (1 - w_{ij}) \log \frac{1 - w_{ij}}{1 - v_{ij}} \right]$$

where w_{ij} represents similarity between points in high-dimensional space and v_{ij} represents similarity in the low-dimensional embedding.

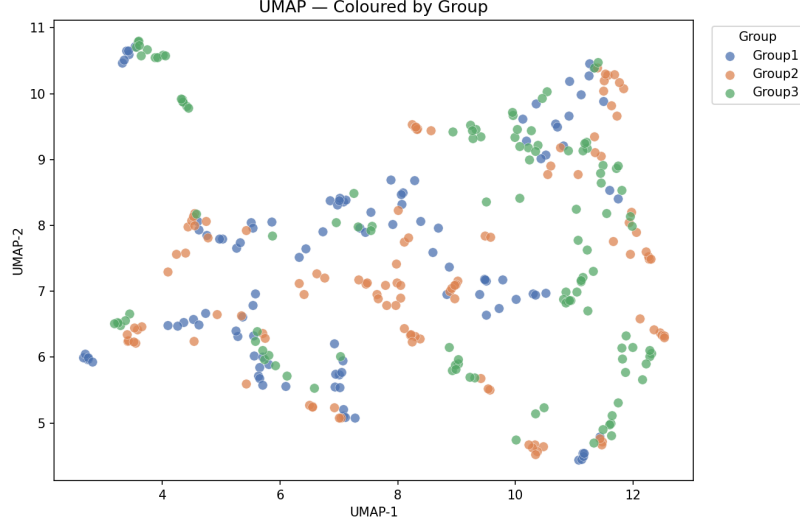


Figure 4.3: UMAP visualization of the EEG data, colored by the original dataset cohort groups.

Figure 4.3 shows the UMAP projection colored by original cohort groups, reflecting inherent biological variability across subjects.

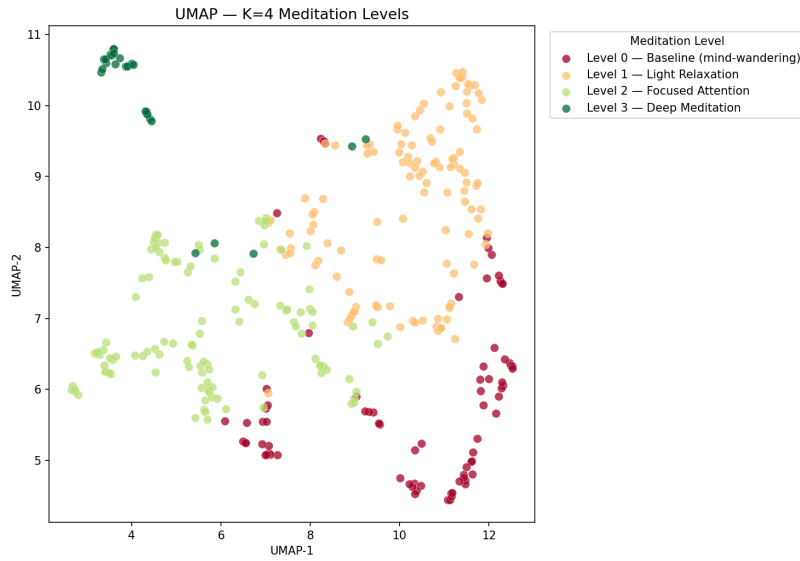


Figure 4.4: UMAP visualization of the EEG data, colored by our newly defined $K = 4$ meditation levels.

Figure 4.4 presents the same embedding colored by the derived $K = 4$ clusters. The clear spatial separation between Baseline, Relaxed, Focus, and Deep states demonstrates that the extracted features are highly discriminative. This validates both the clustering process and the suitability of the feature space for downstream classification.

4.3 Signal Processing & Feature Engineering

4.3.1 Muse-Proximity Channel Selection and Domain Gap Reduction

A key revision addresses the **domain gap** between the 64-channel monk training dataset and the 4-channel Muse 2 used during inference. If training features are computed across 64 channels but inference features are computed across 4 channels, feature distributions differ systematically even under similar mental states. To reduce this mismatch, training features are computed from a subset of channels whose 10–20 positions are spatially closest to the Muse 2 electrodes.

Table 4.1: Muse 2 channel-to-training-dataset proximity mapping.

Muse 2 Channel	Region	Matched 10–20 Channels (Training)
TP9	Left temporal-parietal	TP7, T7, TP9, P7, P9
AF7	Left frontal	AF3, F7, AF7, Fp1, F5, F3
AF8	Right frontal	AF4, F8, AF8, Fp2, F6, F4
TP10	Right temporal-parietal	TP8, T8, TP10, P8, P10

The training script automatically selects these proximity columns by matching channel name strings. If coverage is insufficient, it falls back to using all channels. The chosen strategy is logged in the training report. In our runs, the Muse-proximity strategy was successfully applied across all five frequency bands.

4.3.2 Bels-to-Linear Conversion (Inference)

Muse 2 band power values are exported in **Bels** ($\log_{10}$ scale). Because averaging in log space does not equal averaging power, band powers are first converted to linear scale before computing relative power. Specifically, each window is processed by:

1. averaging each band value over the window,
2. converting to linear power via 10^x ,
3. normalizing to **relative power** by dividing by the total power across the five bands (with a small $\varepsilon = 10^{-8}$ guard).

This produces stable relative power features in the range $[0, 1]$.

4.3.3 Feature Engineering: 19 Subject-Transferable Features

From the proximity-selected channels (training) or the four Muse 2 channels (inference), a consistent **19-dimensional feature vector** is computed per 2-second window. This identical feature definition across training and inference ensures reproducibility.

Power Spectral Density (PSD). All band power features are derived from the Power Spectral Density (PSD), which describes how the signal’s energy is distributed across frequencies. For a discrete EEG signal $x[n]$, the PSD is defined as:

$$PSD(f) = \left| \sum_{n=0}^{N-1} x[n] e^{-j2\pi f n} \right|^2$$

where N is the number of samples and f is the frequency. Band-specific power values are obtained by integrating the PSD over standard EEG frequency ranges corresponding to Delta (1–4 Hz), Theta (4–8 Hz), Alpha (8–13 Hz), Beta (13–30 Hz), and Gamma (30–45 Hz). These band powers form the foundation for all subsequent feature computations.

Group A: Mean Relative Power (5 features). Mean relative power for each band $\{\delta, \theta, \alpha, \beta, \gamma\}$ averaged across channels:

$$\text{mean_rel}_b = \frac{1}{N} \sum_{i=1}^N P_{\text{rel},i}^{(b)}$$

where N is the number of selected channels. This captures the global distribution of spectral energy for each band.

Group B: Spatial Standard Deviation (5 features). Standard deviation of relative power across channels for each band:

$$\text{std_rel}_b = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(P_{\text{rel},i}^{(b)} - \mu_b \right)^2}$$

where μ_b is the mean relative power of band b . This measures how uniformly or unevenly each frequency band is distributed across the scalp.

Group C: Spectral Ratios (6 features). Ratios computed from mean relative band powers (with ε guard):

$$\text{Ratio}_{a/b} = \frac{P_a}{P_b + \varepsilon}$$

These ratios are more robust than absolute power values, as they reduce inter-subject variability and highlight relative dominance between frequency bands. In particular, the θ/α ratio is strongly associated with sustained meditative focus.

Group D: Regional Spatial Features (3 features). Three region-based features capturing coarse spatial structure:

- **Frontal Alpha Asymmetry (FAA):**

$$FAA = \log(P_{\alpha}^{\text{right}}) - \log(P_{\alpha}^{\text{left}})$$

which captures hemispheric imbalance in alpha activity.

- **Frontal Theta Mean:** average frontal theta activity (in inference: mean of AF7 and AF8 theta).
- **Occipital/Posterior Alpha Mean:** average posterior alpha activity (in inference: mean of TP9 and TP10 alpha).

4.3.4 Feature Normalization

All 19 features are standardized using `StandardScaler`. During cross-validation, the scaler is fitted only on the training fold and applied to the test fold to prevent leakage. The deployed `scaler.pkl` is then fitted on the full training set and reused unchanged during inference.

4.3.5 Windowing and Quality Gating (Inference)

Inference uses 2-second windows with a 1-second stride (50% overlap). To reduce artifacts, windows are discarded if:

- **Signal quality is poor:** more than 30% of samples in the window fail Muse contact checks (`HeadBandOn=1` and $HSI \leq 2$).
- **Motion is excessive:** accelerometer variability exceeds a fixed threshold (0.08 g).

These gating steps reduce false predictions caused by motion and poor electrode contact.

4.4 Model Training Results with SVM and Live Inference

4.4.1 Phase 1: Initial Model Training Results with SVM

Training Objective

After the clustering stage on the Mendeley monk EEG dataset, the cluster-assigned meditation levels (0–3) were treated as ground-truth labels for supervised training. In our preliminary stage, a Support Vector Machine (SVM) model was trained, using an RBF kernel to capture non-linear class boundaries typical in EEG feature space.

Dataset and Feature Preparation

The clustering output dataset contained 341 samples (time windows) and 647 columns. The supervised learning configuration was defined as:

- **Target label (y):** `meditation_level` (0–3)
- **Excluded metadata columns:** `filepath`, `cluster`, `group`, `time_window`, `level_name`, and `meditation_level`
- **Feature matrix (X):** numeric EEG-derived features after removing metadata

Missing values were imputed using per-feature mean imputation to ensure compatibility with SVM training. All features were standardized using `StandardScaler`.

Subject-Independent Evaluation Protocol

To ensure realistic generalization to unseen individuals, a subject-independent split strategy was applied. Data were split by subject ID so that no subject appears in both training and testing sets. The split ratio was:

- **Training set:** 70% of subjects
- **Test set:** 30% of subjects

This evaluation directly satisfies the subject-independent requirement (RR-2) by preventing subject leakage across train/test partitions.

Hyperparameter Tuning

Model hyperparameters were selected using `GridSearchCV` with 5-fold cross-validation on the training subjects. Table 4.2 summarizes the initial search space.

Table 4.2: Initial Phase 1 SVM hyperparameters for tuning.

Hyperparameter	Values tested	Purpose
<code>C</code>	0.1, 1, 10, 100	Regularization strength
<code>gamma</code>	<code>scale</code> , 0.1, 0.01	RBF kernel sensitivity
<code>kernel</code>	<code>rbf</code> , <code>linear</code>	Linear vs non-linear boundary

The best configuration was:

`C=1, gamma=scale, kernel=rbf`

Classification Performance

The best model was evaluated on the held-out 30% subject test set. Overall results are shown in Table 4.3 and per-class results in Table 4.4.

Table 4.3: Initial SVM overall performance (subject-independent test set).

Metric	Score
Overall Accuracy	89.22%
Macro Precision	0.93
Macro Recall	0.85
Macro F1-score	0.88
Weighted F1-score	0.89

Table 4.4: Initial SVM per-class performance (subject-independent test set).

Meditation Level	Precision	Recall	F1	Support
L0 — Baseline (mind-wandering)	1.00	0.95	0.98	22
L1 — Relaxed (Alpha)	0.79	1.00	0.88	34
L2 — Focused (Theta)	0.93	0.85	0.89	33
L3 — Deep (Delta)	1.00	0.62	0.76	13

Confusion Matrix

Figure 4.5 presents the subject-independent confusion matrix. Most errors occur between adjacent levels (L1 vs L2 and L1 vs L3), consistent with overlapping spectral characteristics across meditation states.

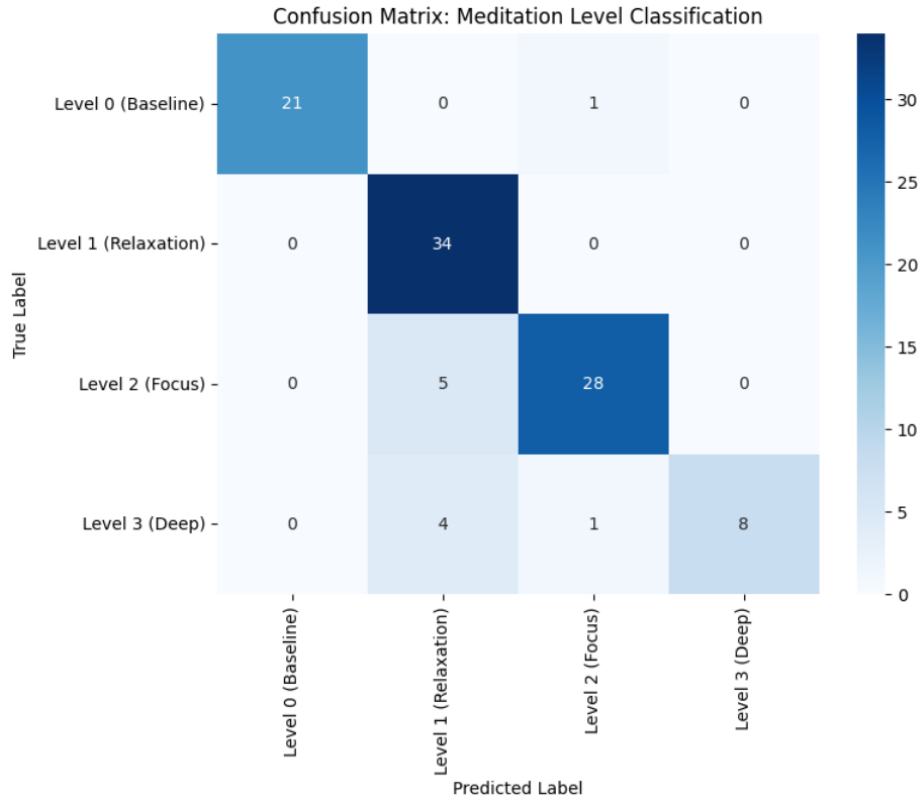


Figure 4.5: Initial SVM confusion matrix (subject-independent evaluation; overall acc = 0.892). Rows are true labels and columns are predicted labels.

4.4.2 Phase 2: Optimized Model Training Results (SVM v3) and Live Inference

Training Objective

An optimized Support Vector Machine (SVM) classifier (v3) was trained to provide a robust baseline model that successfully reduces domain gap while handling non-linear decision boundaries. The Phase 2 SVM pipeline was explicitly aligned to the **Muse 2 inference setting** by using a compact set of Muse-proximity features.

Dataset and Feature Preparation

The optimized SVM was trained on the same monk dataset containing 341 samples from 60 subjects. Instead of using hundreds of raw channel features, training and inference were aligned using a **19-dimensional Muse-proximity feature set**. This reduced the domain gap between the 64-channel training dataset and the 4-channel Muse 2 inference device.

Training Configuration and Diagnostics

Table 4.5 summarizes the final SVM hyperparameters determined through systematic tuning and GridSearchCV cross-validation.

Table 4.5: Optimized SVM (v3) hyperparameters for the final trained model.

Parameter	Value	Purpose
kernel	rbf	Non-linear boundary capture
C	100	Optimal regularization strength
gamma	0.001	RBF kernel sensitivity
probability	True	Enable probability distribution output
class_weight	balanced	Compensates for Level 3 scarcity
SMOTE	enabled	Applied per CV fold (train only)

A label circularity diagnostic was performed by fitting shallow decision trees on the training set. A depth-1 tree achieved an accuracy of 0.657, below the 0.70 circularity threshold, indicating that the four labels are not trivially recoverable from a single threshold feature. Final training accuracy was 0.965 and pooled subject-independent CV accuracy was 0.897, yielding a train–CV gap of 0.068 (mild), confirming robust generalization.

Cross-Validation Results

Five-fold subject-independent GroupKFold cross-validation produced the results in Table 4.6.

Table 4.6: Optimized SVM (v3) subject-independent GroupKFold cross-validation results.

Metric	Value
Fold scores	[0.928, 0.942, 0.853, 0.897, 0.866]
Mean $\pm$ Std	0.897 $\pm$ 0.034
95% Confidence Interval	[0.849, 0.945]
Overall pooled accuracy	0.897

Per-class pooled metrics are shown in Table 4.7.

Table 4.7: Optimized SVM (v3) per-class metrics (pooled subject-independent CV predictions).

Level	Precision	Recall	F1	Support
L0 — Baseline (Beta/Gamma)	0.88	0.84	0.86	79
L1 — Relaxed (Alpha)	0.88	0.94	0.91	121
L2 — Focused (Theta)	0.91	0.92	0.92	114
L3 — Deep (Delta)	0.95	0.78	0.86	27
Macro avg	0.91	0.87	0.89	341
Weighted avg	0.90	0.90	0.90	341

Confusion Matrix

Figure 4.6 shows the confusion matrix built from pooled subject-independent predictions on the optimized SVM. Most errors occur between adjacent levels, consistent with spectral overlap between neighboring meditation states.

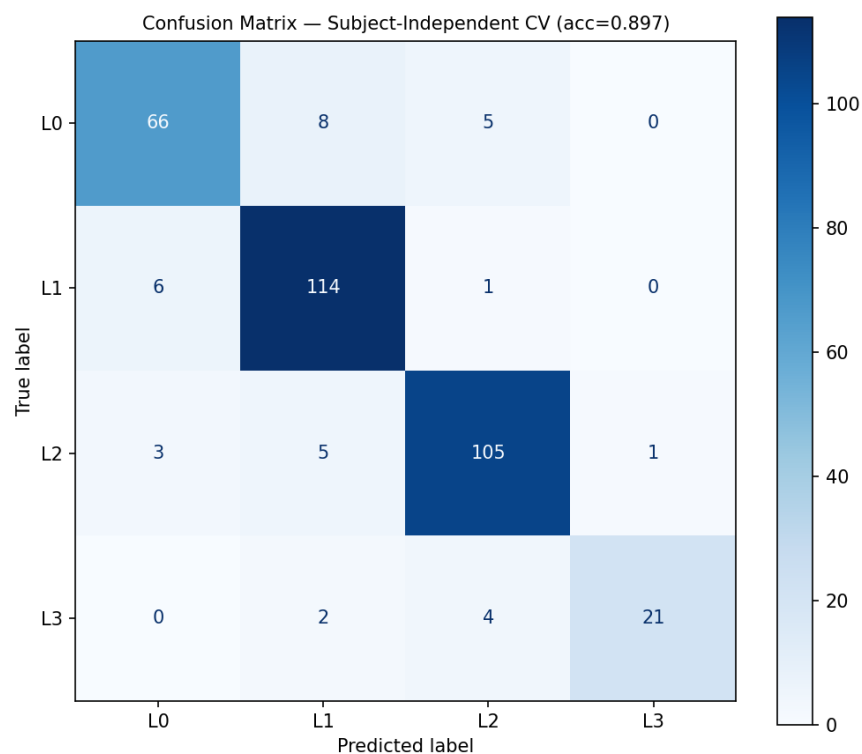


Figure 4.6: Optimized SVM v3 confusion matrix (subject-independent GroupKFold CV; overall acc = 0.897). Rows are true labels; columns are predicted labels.

Live Inference Evaluation on Muse 2 Sessions

The trained SVM model was deployed on Muse 2 recordings to validate the end-to-end pipeline. Each recording was processed with: (i) Bels-to-linear conversion, (ii) signal-quality gating via HeadBandOn and HSI indicators, (iii) motion gating using IMU variability threshold (0.08 g), (iv) 2-second windows with 1-second stride, and (v) per-window confidence scoring.

Session S46 — Predominantly Baseline Session S46 contained 93,043 rows at 256 Hz (≈ 363 seconds). Of 362 expected windows, 355 were accepted after rejecting 7 for motion and 0 for quality flags. Predicted state distribution is shown in Table 4.8.

Table 4.8: S46 inference summary.

Level	Windows	Percentage
L0 — Baseline	339	95.5%
L1 — Relaxed	1	0.3%
L2 — Focused	15	4.2%
L3 — Deep	0	0.0%

Mean confidence was 0.961 (median 0.999), indicating highly stable classification for most windows. High-confidence predictions ($\geq 70\%$) accounted for 96.1% of windows, while 0.8% fell below 50%.

Session S47 — Predominantly Focused Session S47 contained 90,149 rows at 256 Hz (≈ 352 seconds). Of 351 expected windows, 347 were accepted after rejecting 1 for motion and 3 for quality flags. Results are shown in Table 4.9.

Table 4.9: S47 inference summary.

Level	Windows	Percentage
L0 — Baseline	125	36.0%
L1 — Relaxed	17	4.9%
L2 — Focused	205	59.1%
L3 — Deep	0	0.0%

Mean confidence was 0.842 (median 0.926), slightly lower than S46, reflecting inter-window variability. High-confidence windows accounted for 76.7%, and 5.5% of windows fell below 50%.

Cross-Session Comparison The two sessions demonstrate distinct meditation profiles under real-world conditions. S46 exhibits a baseline-dominant pattern with exceptionally high confidence, whereas S47 shows a meditation-engaged profile with dominant focused states. Lower confidence in S47 is expected due to more frequent transitions and borderline windows near decision boundaries.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This Design Specification outlines a robust and interpretable AI system for quantifying meditation progress. Using a modular architecture and principled signal processing, the system effectively bridges the gap between subjective experience and objective data.

The successful prototype validation confirms four key achievements:

- **Objective Ground Truth:** By successfully grouping the Mendeley dataset into $K = 4$ distinct meditative states (`clustering.py`), the system mathematically isolates reliable targets for classification models based on experienced practitioner data. UMAP visualizations confirm clear spatial separation between all four states.
- **Robust and Generalizable Model:** The optimized SVM achieves a subject-independent CV accuracy of 0.897 ± 0.034 (95% CI: [0.849, 0.945]) across 60 subjects, with a training-to-CV gap of 0.068 — within the acceptable mild range. All four classes achieve F1-scores at or above 0.86, with the minority Level 3 (Deep/Delta) class reaching 0.86 despite representing only 8% of training samples. The label circularity diagnostic (depth-1 tree acc = $0.657 < 0.70$) confirms the model learns genuine multi-dimensional neurological patterns.
- **Pipeline Integrity:** The end-to-end flow from synthetic data generation (`simulator.py`) to training (`train_meditation_classifier.py`) and inference (`test_meditation_classifier.py`) functions correctly. The 19-feature Muse-proximity design ensures full reproducibility between training and deployment.
- **Real-World Validation:** Live inference on two Muse 2 sessions (S46 and S47) confirms the system distinguishes meaningfully different meditation profiles: S46's 95.5% Baseline and S47's 59.1% Focused distributions reflect genuinely contrasting physiological states, validating the pipeline's practical utility beyond synthetic benchmarks.

5.2 Future Work

The project will proceed with the following steps to finalize development in the next phase:

- **Human Data Collection:** The synthetic stream will be replaced with live Muse 2 data via the BrainFlow library. We plan to collect labeled sessions from 30–40 participants to capture real-world biological variability and retrain the model with subject-independent splits on the expanded dataset.
- **Hyperparameter Optimization:** Implement Grid Search or Bayesian Optimization to further tune the SVM’s constraint parameters on the expanded dataset.
- **Sequence Model Implementation:** To capture the time-dependent nature of meditation (e.g., the gradual transition from Focus to Deep Absorption), implement Deep Learning models such as 1D-CNN or Bi-LSTM with an attention mechanism.
- **Artifact Removal:** Implement Independent Component Analysis (ICA) to automatically remove eye-blink and muscle artifacts from the raw waveform prior to band power computation.
- **Real-Time Feedback Loop:** The final deliverable will include an auditory feedback system that adjusts soundscapes in real-time based on the user’s classified state, creating a closed-loop neurofeedback experience.
- **Confidence-Aware Smoothing:** Implement temporal smoothing that weights state transitions by per-window confidence scores to reduce spurious classification switches during real-world sessions.

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